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Outrageous Fortune

In the wooded foothills of the Austrian Alps, on the outskirts of Vienna, there is a hut in a vineyard with a small mirror in the window. The vineyard has been there for centuries; it was already old when one of the founders of the Vienna Circle, Otto Neurath, met with Einstein and other scientists on the neighboring hill in 1920 to discuss his idea of an *International Encyclopedia of Unified Science*. The mirror is a newer addition. It was placed there in 2011, by graduate students at the Institute for Quantum Optics and Quantum Information of the University of Vienna, as part of a long-distance quantum encryption network. The students, working for Professor Anton Zeilinger, shot photons at the mirror, one at a time, from their lab on the top floor of their building in the heart of Vienna, five kilometers away. On the roof of that building, a specially equipped telescope—named after Hedy Lamarr, the Viennese movie star and cryptography pioneer—remained trained on the mirror in the vineyard, carefully collecting the reflected packets of light after their passage through the turbulent air over Vienna.

This feat, unimaginable outside of thought experiments for the founders of quantum theory, was merely a test. Right now, Zeilinger and his students are using this same equipment to exchange photons with a specially designed satellite in low Earth orbit, attempting to enable quantum-encrypted communication between Vienna and Yunnan Astronomical Observatory in China, where physicist Jianwei Pan, a former student of Zeilinger, has a similar setup already in place. And if the past is any guide,